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The principles of dust emission monitoring: Part 2

June 2005, Analytical Instrumentation & Environmental Monitoring

In our **last issue** we looked at mass concentration, mass flow and total mass emissions. In Part 2 we begin with indicative and quantitative monitoring.

There is one further method of reporting dust emissions which is unique to the measurement of dust and, indeed, unique to the United Kingdom. This is the principle of indicative monitoring. All the preceding discussion has involved the use of quantitative monitoring for the absolute determination of levels of pollution emission.

Indicative monitors present data that is not calibrated in any units of mass emission. Such devices are expected to be proportional to mass emission in some way, so that they can be used to indicate to an operator whether the process dust emissions are at normal or abnormal levels.

An example of such a use might be on the output of a fabric bag filter where dust emissions would be expected to be low. An indicative monitor would be used as a bag failure alarm to indicate a tear in a bag, releasing abnormally high levels of dust. The monitor would not, however, be able to report how much dust was being released but would simply provide an alarm to the operator that something has gone wrong.

What exactly is the difference, therefore, between an indicative monitor and a quantitative monitor? In so far as any measurement device must be uniquely responsive to the parameter to be measured, both indicative and quantitative monitors must fall within that definition.

It may be that an indicative monitor does not require the same degree of accuracy as a quantitative analyser, but it should be noted that the question of measurement accuracy has not previously arisen in the discussion of these principles. An instrument should be sufficiently accurate for the range it is measuring.

If any sensor cannot be shown to respond uniquely to dust levels, it can be used neither as a quantitative nor an indicative measurement. The essential difference between an indicative sensor and a quantitative sensor is purely one of calibration. The only practical method of calibrating a dust monitor is by comparison against a series of isokinetic sampling tests. Prior to such tests the dust sensor is purely indicative. Subsequent to this calibration the sensor may be defined as quantitative. This then becomes the defining difference between indicative and quantitative monitors. Quantitative are calibrated, indicative are not.

In so far as any process is allowed to utilise indicative sensors, there is usually an obligation to perform an absolute isokinetic test on the emission at least once per annum. It would surely make sense to use that data to calibrate the indicative sensor and turn it into a quantitative sensor, without significant further cost to industry.

Dust monitor calibration

All dust monitoring devices, with the possible exception of beta-gauge devices are inferential devices. That is, they infer the levels of dust emission from the measurement of another parameter that is presumed to be proportional to the dust content of the exhaust gases.

Unlike a gas analyser, which can be calibrated by injecting gas of a known concentration, dust monitors cannot be challenged by a fixed level of dust. The basic problem is that while one molecule of, say, carbon monoxide, behaves the same as any other molecule of carbon monoxide, no two particles of dust are the same. There is no such concept as standard dust, as there is with standard gas.

The only absolute method of calibrating a dust sensor is to compare it with data taken from an isokinetic dust sampling test, in which dust is collected on a filter from a fixed volume of gas drawn from the exhaust gases. The weight of dust collected and the volume of gas drawn enables a mass concentration and indeed mass flow to be calculated.

Auto Zero and Span Measurement verification is a key issue in environmental monitoring. Gas analysers are expected to have provision built in to enable them to be challenged with zero and span gas either manually or automatically at periodic intervals to ensure that their calibration is correct. Whilst it would perhaps be desirable to have the same facility in a dust monitor, it is in practice impossible to achieve.

One of the major errors in dust measurement is derived from the way in which the mass of dust is inferred from the measured parameter. Optical instruments, for example, measure the light scattering properties of particles, properties that vary considerably with material type and particle size. Even if it were possible to devise a 'standard dust' and inject it into the analyser measurement path, it would not necessarily bear any resemblance to how the analyser behaves towards the dust in the stack at any one time.

Additionally, because of the intrinsic problems associated with continuous isokinetic sampling, all dust

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monitors tend to be of an in situ nature. As a result, to introduce a zero check on an in situ measurement requires significant compromises in the way in which zero is checked. Since the monitors are measuring the dust content in situ it is not conceivable to produce a zero dust environment for the analyser. This invariably means that the zero check is at best limited in its abilities to test all sources of zero error and worst, downright wrong.

The nearest any legislative authority has come to demanding auto zero and span checks is the US EPA which requires them on opacity monitors used within the Acid Rain Programme (CFR 40 Part 75). The point is, however, that Part 75 requires the reporting of opacity, not dust, so that the question of how the measured opacity relates to differing types of dust is not an issue.

It would be extremely counter-productive to impose the same zero and span checking facilities on dust measurements as now exists on gas measurement. The two sets of measurement are different in that dust is an inferential and gas a direct measurement.

Dust measurement techniques

There are two main competing technologies for the determination of levels of dust emissions. These are optical systems and tribo-electric systems. Within these two groups are a number of different types of instruments, all of which show similar operating characteristics.

Optical measurement

Historically, the earliest form of dust sensor was the simple optical smoke monitor, which effectively measured the opacity of the stack. Since then, more sophisticated forms of opacity monitor have been introduced along with a variety of light scattering sensors. All these instruments, however, share a number of common features.

1. All optical sensors respond to the mass concentration of dust within the duct.

Light beams are influenced by the scattering characteristics of large numbers of small particles and the measured parameter is a function of the number of particles per unit volume of gas, which in turn is proportional to the mass concentration.

2. All optical sensors are crucially dependent on the scattering characteristics of the particles in the gas stream. In particular, the size and material of the particles will have a major influence on the way in which light is scattered. One of the main criticisms of traditional opacity monitors is that the resultant mass concentration calibration varies considerably with particulate type.

Optical backscatter and forward scatter devices, introduced to enable lower density levels of emission to be measured, suffer especially from particle size and material variations.

Tribo-electric sensors

Tribo-electric sensors fall into two categories - DC sensors and AC sensors. Both types are fundamental sensors of mass flow and can be calibrated directly in units of gm/sec or kg/hr. DC sensors have been shown to have a linear response to mass flow over a wide range of gas velocities. These devices provide a simple low-cost mass flow measurement that, unlike mass concentration measurement, requires no further normalisation to deliver an absolute measurement of environmental impact.

However, AC devices, because of their additional AC coupling, are also dependent for their signal on the degree of inhomogeneity of the dust within the gas flow. This imbues them with unpredictable velocity sensitivity, making them unsuitable for any quantitative analysis, unless the process conditions and exhaust gas velocities are absolutely stable. Further measurement of flow is of no help since the flow dependency will vary from plant to plant and from location to location within even the same duct, because of the effects of gas turbulence patterns on the inhomogeneity of the dust flow.

Conclusions

Mass concentration reports are only of use when they can be related to a defined standard. That is, the data must be normalised to standard values of temperature, pressure, oxygen and water vapour. In the case of non-combustion processes where there is no method to measure the dilution of the emissions, mass concentration is almost worthless as a reporting standard. This creates serious cost and feasibility issues for the reporting of dust emissions in mass concentration from Schedule B processes.

Mass flow, however, is an absolute standard. It provides a measurement of total mass emission from a source and may be used as a direct assessment of environmental impact. It is valid for any type of process, combustion or non-combustion and may be considered as being the most relevant reporting method for Schedule B.

Of the main measurement techniques, optical sensors are mass concentration sensitive devices, while tribo-electric probes are proportional to mass flow. In particular, the DC tribo-electric probe is linearly proportional to mass flow and thus offers a simple low-cost solution to dust emission monitoring, by providing a direct measurement of the environmental impact of any emission source.

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